CITY EMERGENCY COMMAND CENTER

A smart city operates a central emergency command center that receives crisis alerts from different zones in real time. Due to limited manpower, the system cannot process all alerts at once and must decide how to prioritize and manage them efficiently.

Each alert contains:

- Zone ID
- Emergency type (fire, flood, accident, power failure, etc.)
- Severity level (1–10)
- Timestamp of arrival

Situation:

Alerts keep arriving continuously from multiple zones. The command center must always focus on the **most recent critical situation first**, but also ensure no important alert is ignored.

Student Activity (Group of 3):

Students must **design the complete stack-based solution (not code)** and answer:

1. How will you represent each alert in memory? (structure design)
2. Why is stack suitable for handling emergency alerts in this system?
3. How will you insert new alerts?
4. How will you decide which alert to process next?
5. What happens if a higher severity alert arrives after several are already stored?
6. How will you find:
   - Highest severity alert
   - Oldest pending alert
7. Draw a **step-by-step simulation** of at least 6 alert insertions and 3 removals.
8. Suggest one real-life improvement to this system.

Twist :

- Sometimes a “national emergency” alert arrives — how will your design handle it?
- Can stack alone fully solve fairness? Justify.